

Understand what is the overall design of the computer system:

1. Input
2. Process
3. Output
4. Storage

Understand what is the meaning of a System

- A system is a set of interconnected components that work together to achieve a common goal or purpose.
- **Key Characteristics of System**
 - Purpose/Goal
 - Components/elements/subsystem
 - Interconnections/Relationships amongst the components
 - Boundary
 - People
 - Input and Output
 - Processes
 - Environment
 - System is part of the environment and interact with it
 - **Environment Elements**
 - **Users**
 - **External Systems**
 - Other Systems
 - APIs
 - **Physical Infrastructure/Hardware**
 - Servers
 - Networks
 - Computers
 - Devices
 - Sensors
 - **Platforms**
 - Operating Systems
 - third party software
 - databases
 - cloud services
 - **Policies**
 - Data Protection : GDPR
 - Intellectual Property Laws

- **Data**
 - **Physical Environment**
 - Electricity
 - Internet Connectivity
 - Natural Events
 - **Organizations/Stakeholders**
- **Relationships amongst those terms:**
- **Component**
 - Feature
 - **Process**
 - **Operation**
- **Component Elements**
 - Responsibilities
 - Dependencies
 - interfaces
 - Resources
 - Constraints
 - Features
 - **Feature 1**
 - **Process**
 - **Operations**
 - **Feature 2**
 - **Process 1**
 - **Operation 1**
 - **Operation 2**
 - **Process 2**
 - **Process 3**
- **Process Elements:**
 - Input
 - Steps
 - Control/ Logic
 - Resources
 - Output
- **Operation Elements**
 - Input parameters
 - Function logic or calculations
 - Output/return value

System Analysis Framework (OOA-focused)

1. **Determine the main system**
 - Identify the overall system that needs to be analyzed and developed.
 - Define the system's scope, objectives, and boundaries.

2. **Define the system overall statement**
 - Write a clear, high-level description of the system's purpose, goals, and main functions.
 - Include key constraints, stakeholders, and expected outcomes.
3. **Divide the system into subsystems**
 - Break down the system into major functional modules or areas.
 - Assign responsibilities to each subsystem.
4. **Select one subsystem and further divide it into smaller components or features**
 - Decompose subsystems into smaller units for detailed analysis.
 - Identify individual features or modules that can be implemented separately.
5. **Analyze the requirements of each component**
 - Gather functional requirements (what the component should do) and non-functional requirements (performance, security, usability, etc.).
 - Identify constraints, business rules, and dependencies.
6. **Determine the users of the system**
 - Identify all actors interacting with the system (human users, external systems, devices).
 - Define their roles, goals, and level of access.
7. **Model the requirements using use case diagrams**
 - Represent the interactions between actors and the system.
 - Capture primary use cases, alternative flows, and exceptions.
8. **Develop scenarios and sequence diagrams (optional but recommended)**
 - Write detailed scenarios for each use case.
 - Create sequence diagrams to visualize the flow of operations and interactions between objects over time.
9. **Determine the main entities (classes) in the system and their attributes and behaviors**
 - Identify objects/classes, their properties (attributes), and methods (behaviors).
 - Group related behaviors and attributes logically.
10. **Define relationships between classes**
 - Show associations (connections), dependencies, generalizations (inheritance), and aggregations/compositions (whole-part relationships).
 - Identify multiplicities (one-to-one, one-to-many, many-to-many).
11. **Create class diagrams**
 - Visually represent classes, attributes, methods, and relationships.
 - Ensure diagrams reflect real-world scenarios and the previously defined use cases.
12. **Refine and iterate the models**
 - Review the diagrams and requirements with stakeholders.
 - Resolve conflicts, clarify ambiguities, and ensure completeness.
 - Update use case diagrams, class diagrams, and relationships as needed.
13. **Validate the system model with stakeholders**
 - Confirm that the analysis accurately captures the user requirements and business needs.
 - Use walkthroughs, reviews, or prototyping to get feedback.
14. **Document the analysis**

- Create a comprehensive system analysis document including:
 - Overall system statement
 - Subsystems and components
 - Requirements (functional & non-functional)
 - Use case diagrams and scenarios
 - Class diagrams and relationships
 - Assumptions and constraints